Importing all the required libraties

In [81]:
```
import numpy as np
import pandas as pd
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

Loading, Pre-Processing data

In [82]:
```
df = pd.read_csv('full_german_credit.csv')
df.head()
```

Out[82]:

	Status_of_existing_checking_account	Duration_in_month	Credit_history	Purpos
0	< 0 DM	6	critical account / other credits existing (not...	radio/T
1	0 <= ... < 200 DM	48	existing credits paid back duly till now	radio/T
2	no checking account	12	critical account / other credits existing (not...	educatic
3	< 0 DM	42	existing credits paid back duly till now	furniture/equipmer
4	< 0 DM	24	delay in paying off in the past	car (nev

5 rows × 21 columns

In [83]:
```
print(df.info())
print(df.nunique())
```

```
print(df.notnull())
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 21 columns):
 #   Column                                                Non-Null Count  Dtype
---  ------                                                --------------  -----
 0   Status_of_existing_checking_account                   1000 non-null   objec
t
 1   Duration_in_month                                     1000 non-null   int64
 2   Credit_history                                        1000 non-null   objec
t
 3   Purpose                                               1000 non-null   objec
t
 4   Credit_amount                                         1000 non-null   int64
 5   Savings_account_bonds                                 1000 non-null   objec
t
 6   Present_employment_since                              1000 non-null   objec
t
 7   Installment_rate_in_percentage_of_disposable_income   1000 non-null   int64
 8   Personal_status_and_sex                               1000 non-null   objec
t
 9   Other_debtors_guarantors                              93 non-null     objec
t
 10  Present_residence_since                               1000 non-null   int64
 11  Property                                              1000 non-null   objec
t
 12  Age_in_years                                          1000 non-null   int64
 13  Other_installment_plans                               186 non-null    objec
t
 14  Housing                                               1000 non-null   objec
t
 15  Number_of_existing_credits_at_this_bank               1000 non-null   int64
 16  Job                                                   1000 non-null   objec
t
 17  Number_of_people_being_liable_to_provide_maintenance_for  1000 non-null   int64
 18  Telephone                                             404 non-null    objec
t
 19  Foreign_worker                                        1000 non-null   objec
t
 20  Risk                                                  1000 non-null   objec
t
dtypes: int64(7), object(14)
memory usage: 164.2+ KB
None
Status_of_existing_checking_account                          4
Duration_in_month                                           33
Credit_history                                               5
Purpose                                                     10
Credit_amount                                              921
Savings_account_bonds                                        5
Present_employment_since                                     5
Installment_rate_in_percentage_of_disposable_income          4
Personal_status_and_sex                                      4
Other_debtors_guarantors                                     2
Present_residence_since                                      4
Property                                                     4
Age_in_years                                                53
```

```
Other_installment_plans                                  2
Housing                                                  3
Number_of_existing_credits_at_this_bank                  4
Job                                                      4
Number_of_people_being_liable_to_provide_maintenance_for 2
Telephone                                                1
Foreign_worker                                           2
Risk                                                     2
dtype: int64
     Status_of_existing_checking_account  Duration_in_month  Credit_history  \
0                                   True               True            True
1                                   True               True            True
2                                   True               True            True
3                                   True               True            True
4                                   True               True            True
..                                   ...                ...             ...
995                                 True               True            True
996                                 True               True            True
997                                 True               True            True
998                                 True               True            True
999                                 True               True            True

     Purpose  Credit_amount  Savings_account_bonds  Present_employment_since  \
0       True           True                   True                      True
1       True           True                   True                      True
2       True           True                   True                      True
3       True           True                   True                      True
4       True           True                   True                      True
..       ...            ...                    ...                       ...
995     True           True                   True                      True
996     True           True                   True                      True
997     True           True                   True                      True
998     True           True                   True                      True
999     True           True                   True                      True

     Installment_rate_in_percentage_of_disposable_income  \
0                                                   True
1                                                   True
2                                                   True
3                                                   True
4                                                   True
..                                                   ...
995                                                 True
996                                                 True
997                                                 True
998                                                 True
999                                                 True

     Personal_status_and_sex  Other_debtors_guarantors  ...  Property  \
0                       True                     False  ...      True
1                       True                     False  ...      True
2                       True                     False  ...      True
3                       True                      True  ...      True
4                       True                     False  ...      True
..                       ...                       ...  ...       ...
995                     True                     False  ...      True
```

```
996                    True                     False  ...      True
997                    True                     False  ...      True
998                    True                     False  ...      True
999                    True                     False  ...      True

     Age_in_years  Other_installment_plans  Housing  \
0            True                    False     True
1            True                    False     True
2            True                    False     True
3            True                    False     True
4            True                    False     True
..            ...                      ...      ...
995          True                    False     True
996          True                    False     True
997          True                    False     True
998          True                    False     True
999          True                    False     True

     Number_of_existing_credits_at_this_bank   Job  \
0                                       True  True
1                                       True  True
2                                       True  True
3                                       True  True
4                                       True  True
..                                       ...   ...
995                                     True  True
996                                     True  True
997                                     True  True
998                                     True  True
999                                     True  True

     Number_of_people_being_liable_to_provide_maintenance_for  Telephone  \
0                                                        True       True
1                                                        True      False
2                                                        True      False
3                                                        True      False
4                                                        True      False
..                                                        ...        ...
995                                                      True      False
996                                                      True       True
997                                                      True      False
998                                                      True       True
999                                                      True      False

     Foreign_worker  Risk
0              True  True
1              True  True
2              True  True
3              True  True
4              True  True
..              ...   ...
995            True  True
996            True  True
997            True  True
998            True  True
999            True  True
```

```
[1000 rows x 21 columns]
       Duration_in_month  Credit_amount  \
count        1000.000000    1000.000000
mean           20.903000    3271.258000
std            12.058814    2822.736876
min             4.000000     250.000000
25%            12.000000    1365.500000
50%            18.000000    2319.500000
75%            24.000000    3972.250000
max            72.000000   18424.000000

       Installment_rate_in_percentage_of_disposable_income  \
count                                          1000.000000
mean                                              2.973000
std                                               1.118715
min                                               1.000000
25%                                               2.000000
50%                                               3.000000
75%                                               4.000000
max                                               4.000000

       Present_residence_since  Age_in_years  \
count              1000.000000   1000.000000
mean                  2.845000     35.546000
std                   1.103718     11.375469
min                   1.000000     19.000000
25%                   2.000000     27.000000
50%                   3.000000     33.000000
75%                   4.000000     42.000000
max                   4.000000     75.000000

       Number_of_existing_credits_at_this_bank  \
count                              1000.000000
mean                                  1.407000
std                                   0.577654
min                                   1.000000
25%                                   1.000000
50%                                   1.000000
75%                                   2.000000
max                                   4.000000

       Number_of_people_being_liable_to_provide_maintenance_for
count                                          1000.000000
mean                                              1.155000
std                                               0.362086
min                                               1.000000
25%                                               1.000000
50%                                               1.000000
75%                                               1.000000
max                                               2.000000
```

In [84]:
```
df['Telephone'].fillna('unknown', inplace=True)
df['Other_installment_plans'].fillna('unknown', inplace=True)
df['Other_debtors_guarantors'].fillna('unknown', inplace=True)
```

```
column = df.columns

le = LabelEncoder()

print((df[column[1]].dtypes))

for cols in column:
    if df[cols].dtypes == "object":
        print("entered")
        df['encoded'+cols] = le.fit_transform(df[cols])
```

```
int64
entered
entered
entered
entered
entered
entered
entered
entered
entered
entered
entered
entered
entered
entered
entered
```

```
C:\Users\Aarya Bhave\AppData\Local\Temp\ipykernel_28024\859361229.py:1: FutureWarnin
g: A value is trying to be set on a copy of a DataFrame or Series through chained as
signment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because
the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method
({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform
the operation inplace on the original object.


  df['Telephone'].fillna('unknown', inplace=True)
C:\Users\Aarya Bhave\AppData\Local\Temp\ipykernel_28024\859361229.py:2: FutureWarnin
g: A value is trying to be set on a copy of a DataFrame or Series through chained as
signment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because
the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method
({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform
the operation inplace on the original object.


  df['Other_installment_plans'].fillna('unknown', inplace=True)
C:\Users\Aarya Bhave\AppData\Local\Temp\ipykernel_28024\859361229.py:3: FutureWarnin
g: A value is trying to be set on a copy of a DataFrame or Series through chained as
signment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because
the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method
({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform
the operation inplace on the original object.


  df['Other_debtors_guarantors'].fillna('unknown', inplace=True)
```

In [85]:
```
df.head(2)
```

Out[85]:

	Status_of_existing_checking_account	Duration_in_month	Credit_history	Purpose	Credit_
0	< 0 DM	6	critical account / other credits existing (not...	radio/TV	
1	0 <= ... < 200 DM	48	existing credits paid back duly till now	radio/TV	

2 rows × 35 columns

In [86]:
```
for cols in column:
    if df[cols].dtypes == "object":
        df.drop([cols], axis=1, inplace=True)
```

In [ ]:

In [ ]:

Defining the independent, dependent variables

In [87]:
```
X = df.drop(['encodedRisk'], axis=1)
y = df['encodedRisk']
```

Applying train-test-split

In [88]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=3, shuffle=T
```

Building the network

In [89]:
```
model = Sequential()

model.add(Dense(16, input_dim = X_train.shape[1], activation = 'relu'))
model.add(Dense(10, activation = 'relu'))
model.add(Dense(1, activation='sigmoid'))

model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

```
c:\Users\Aarya Bhave\AppData\Local\Programs\Python\Python313\Lib\site-packages\keras
\src\layers\core\dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim`
argument to a layer. When using Sequential models, prefer using an `Input(shape)` ob
ject as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Training the model

In [90]:
```
train = model.fit(X_train, y_train, epochs=50, batch_size=16, validation_split=0.2)
```

```
Epoch 1/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/step - accuracy: 0.2719 - loss: 383.7868 - val_ac
curacy: 0.4062 - val_loss: 230.5399
Epoch 2/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.2719 - loss: 224.9819 - val_acc
uracy: 0.4062 - val_loss: 118.9230
Epoch 3/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.2750 - loss: 87.9720 - val_accu
racy: 0.4750 - val_loss: 8.3698
Epoch 4/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6750 - loss: 5.1078 - val_accur
acy: 0.5938 - val_loss: 1.9101
Epoch 5/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7141 - loss: 0.9841 - val_accur
acy: 0.6500 - val_loss: 0.9703
Epoch 6/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7234 - loss: 0.8367 - val_accur
acy: 0.6375 - val_loss: 0.9075
Epoch 7/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7078 - loss: 0.7632 - val_accur
acy: 0.6750 - val_loss: 0.8292
Epoch 8/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7250 - loss: 0.7265 - val_accur
acy: 0.6812 - val_loss: 0.7984
Epoch 9/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7031 - loss: 0.7797 - val_accur
acy: 0.6375 - val_loss: 0.7709
Epoch 10/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6844 - loss: 0.9051 - val_accur
acy: 0.6375 - val_loss: 0.6987
Epoch 11/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6828 - loss: 0.9216 - val_accur
acy: 0.6875 - val_loss: 0.7729
Epoch 12/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6438 - loss: 1.2396 - val_accur
acy: 0.6438 - val_loss: 0.9115
Epoch 13/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6734 - loss: 0.9015 - val_accur
acy: 0.6250 - val_loss: 0.9643
Epoch 14/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6656 - loss: 0.8784 - val_accur
acy: 0.6375 - val_loss: 0.6744
Epoch 15/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6922 - loss: 0.6894 - val_accur
acy: 0.6875 - val_loss: 0.6918
Epoch 16/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7000 - loss: 0.6861 - val_accur
acy: 0.6250 - val_loss: 0.6999
Epoch 17/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7172 - loss: 0.6257 - val_accur
acy: 0.6250 - val_loss: 0.7018
Epoch 18/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6969 - loss: 0.7205 - val_accur
acy: 0.6250 - val_loss: 0.7984
Epoch 19/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7297 - loss: 0.6399 - val_accur
```

```
acy: 0.6875 - val_loss: 0.6195
Epoch 20/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7000 - loss: 0.7776 - val_accur
acy: 0.6250 - val_loss: 0.7891
Epoch 21/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7063 - loss: 0.7203 - val_accur
acy: 0.6000 - val_loss: 1.1116
Epoch 22/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.6625 - loss: 0.8532 - val_accur
acy: 0.6125 - val_loss: 0.7851
Epoch 23/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7047 - loss: 0.6603 - val_accur
acy: 0.5938 - val_loss: 1.1966
Epoch 24/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6562 - loss: 1.1066 - val_accur
acy: 0.5938 - val_loss: 1.9737
Epoch 25/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6578 - loss: 1.0055 - val_accur
acy: 0.6313 - val_loss: 0.9207
Epoch 26/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6828 - loss: 0.7759 - val_accur
acy: 0.6938 - val_loss: 0.6451
Epoch 27/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7047 - loss: 0.6550 - val_accur
acy: 0.6875 - val_loss: 0.6159
Epoch 28/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7063 - loss: 0.6805 - val_accur
acy: 0.6438 - val_loss: 0.6575
Epoch 29/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6969 - loss: 0.7579 - val_accur
acy: 0.6812 - val_loss: 0.5983
Epoch 30/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7156 - loss: 0.6305 - val_accur
acy: 0.6125 - val_loss: 0.8120
Epoch 31/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6906 - loss: 0.6660 - val_accur
acy: 0.6125 - val_loss: 0.7594
Epoch 32/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6625 - loss: 1.0204 - val_accur
acy: 0.5938 - val_loss: 1.1379
Epoch 33/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.6891 - loss: 0.7723 - val_accur
acy: 0.6125 - val_loss: 0.7275
Epoch 34/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6656 - loss: 0.8294 - val_accur
acy: 0.6562 - val_loss: 0.5908
Epoch 35/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6984 - loss: 0.6327 - val_accur
acy: 0.5938 - val_loss: 1.2430
Epoch 36/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7156 - loss: 0.6597 - val_accur
acy: 0.6125 - val_loss: 0.8160
Epoch 37/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6906 - loss: 0.8639 - val_accur
acy: 0.5938 - val_loss: 1.4543
Epoch 38/50
```

```
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6875 - loss: 0.7665 - val_accur
acy: 0.6062 - val_loss: 0.9388
Epoch 39/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7359 - loss: 0.5602 - val_accur
acy: 0.6500 - val_loss: 0.5800
Epoch 40/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6984 - loss: 0.6070 - val_accur
acy: 0.6625 - val_loss: 0.6224
Epoch 41/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7266 - loss: 0.5862 - val_accur
acy: 0.6938 - val_loss: 0.5795
Epoch 42/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6781 - loss: 0.6920 - val_accur
acy: 0.6062 - val_loss: 0.8476
Epoch 43/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6938 - loss: 0.8156 - val_accur
acy: 0.6000 - val_loss: 1.0693
Epoch 44/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.6641 - loss: 1.0161 - val_accur
acy: 0.6062 - val_loss: 0.8509
Epoch 45/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7156 - loss: 0.6903 - val_accur
acy: 0.6375 - val_loss: 0.6623
Epoch 46/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7203 - loss: 0.6277 - val_accur
acy: 0.6062 - val_loss: 0.8209
Epoch 47/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7250 - loss: 0.6394 - val_accur
acy: 0.7063 - val_loss: 0.5936
Epoch 48/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.6687 - loss: 0.9359 - val_accur
acy: 0.6000 - val_loss: 0.8756
Epoch 49/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.7141 - loss: 0.6104 - val_accur
acy: 0.6062 - val_loss: 0.8371
Epoch 50/50
40/40 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7156 - loss: 0.6539 - val_accur
acy: 0.6187 - val_loss: 0.6778
```

Calculating the accuracy

In [91]:
```
loss, accuracy = model.evaluate(X_test, y_test)
y_predict = model.predict(X_test)
```

```
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.6900 - loss: 0.6335
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step
```

In [93]:
```
print("Accuracy : ",accuracy)
```

```
Accuracy :  0.6899999976158142
```

In [ ]:
```

```